

25) Given $v \neq 0$ and p in $\mathbb{R}^n$, the line through p in the direction of v has the parametric equation $x = p + tv$. Show that a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ maps this line onto another line or onto a single point (a degenerate line).

Solution: $x = p + tv$. $T(x) = T(p + tv) = T(p) + T(tv)$ (Condition 1)
 $= T(p) + tT(v)$ (Condition 2)

$T(x)$ is the equation of a line through $T(p)$ in the direction of $T(v)$.
 If $t=0$, then $T(x)$ maps to the single point $T(p)$. $\square$

26) Let u and v be linearly independent vectors in $\mathbb{R}^3$, and let P be the plane through u, v and 0 . The parametric equation of P is $x = su + tv$ (with s, t in $\mathbb{R}$). Show that a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ maps P onto a plane through 0 , or onto a line through 0 , or onto just the origin in $\mathbb{R}^3$. What must be true about $T(u)$ and $T(v)$ in order for the image of the plane P to be a plane?

Solution: $T(x) = T(su + tv) = T(su) + T(tv)$ (Condition 1) $= sT(u) + tT(v)$ (Condition 2)

For $T(x)$ to be a plane through 0 , $T(u)$ and $T(v)$ must be linearly independent. Suppose $T(u)$ and $T(v)$ are linearly dependent. $\therefore$ One must be a scalar multiple of the other. WLOG, let $T(v)$ is a scalar multiple of $T(u)$. $\therefore T(v) = aT(u)$ for some $a \in \mathbb{R}$.

$\therefore T(x) = sT(u) + t(aT(u)) = T(u)(s + at) + 0$.

$T(x)$ is then a line through 0 . $\square$
 If $T(u)$ and $T(v)$ are both 0 , then $T(x) = 0 \rightarrow$ origin in $\mathbb{R}^3$. $\square$

27) a) Show that the line through vectors p and q in $\mathbb{R}^n$ may be written in the parametric form $x = (1-t)p + tq$.

Solution: The line through vectors p and q must be parallel to $(q-p)$.
 $\therefore$ The equation of the line is: $p + (q-p)t$
 $= p + (q + (-1) \cdot p) \cdot t$ (Algebraic property vi)
 $= p + ((-1) \cdot p) \cdot t + qt$ (Algebraic property (i)) $= p + (-1) \cdot (p \cdot t) + qt$ (Algebraic property 7)
 $= p + (-1) \cdot (t \cdot p) + qt$ (Algebraic property 7)
 $= p + (-1 \cdot t) \cdot p + qt$ (Algebraic property 2)